

Predicting Airline Passenger Satisfaction

Machine Learning and Deep Learning: A Comparative Evaluation

15,000 passenger records | 14 models compared | B.Tech CSE Research Project

0.9594

Best test accuracy
Stacking Ensemble

0.9627

Best test F1
primary selection metric

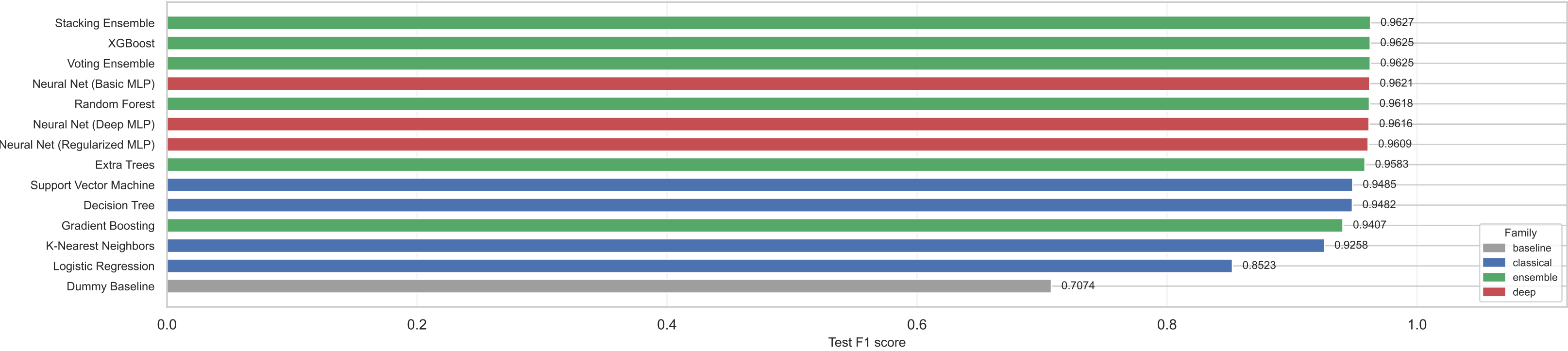
8

Models statistically tied
overlapping 95% CIs

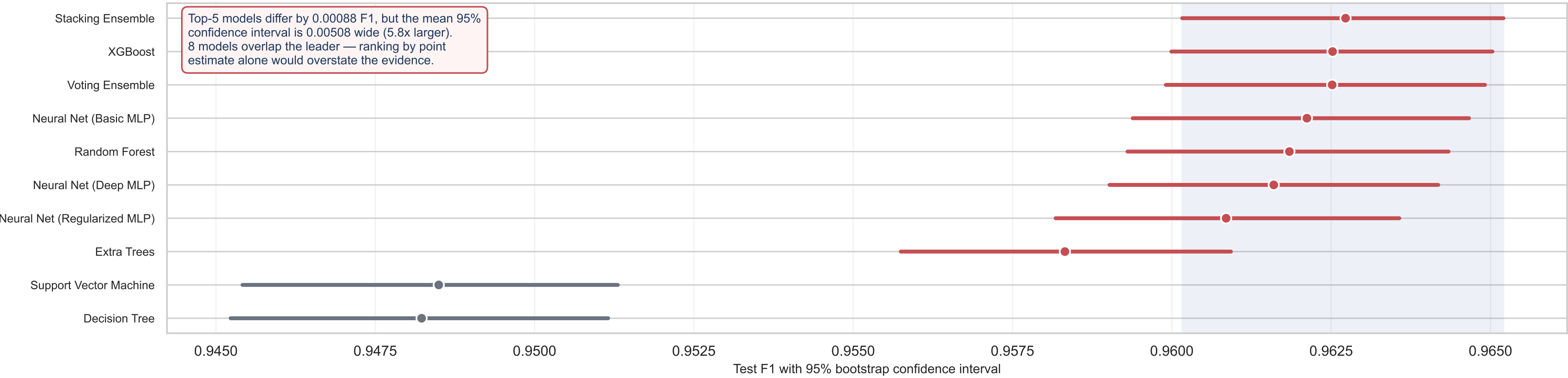
5.8x

CI width vs top-5 spread
uncertainty dominates

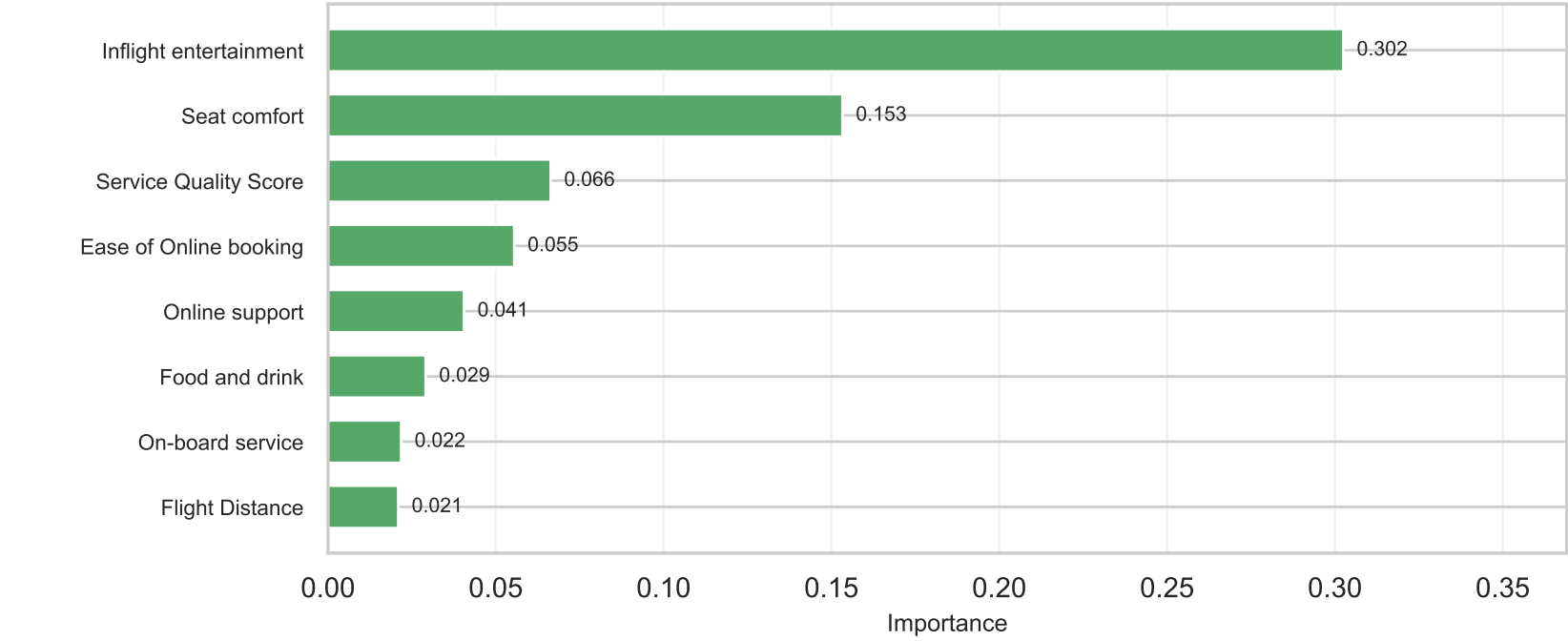
1. Model comparison — 14 models on the held-out test set



2. KEY FINDING — the leading models are statistically indistinguishable



3. What drives the predictions



4. Performance vs training cost



5. Methodology

Split	Stratified 70/15/15 – 10,500 train, 2,250 validation, 2,250 test
Protocol	Test set opened exactly ONCE, after model selection was complete on the validation set
Leakage	All preprocessing fitted inside sklearn Pipelines on training folds only; disjointness asserted by tests
Tuning	RandomizedSearchCV, 3 candidates x 2 folds (train only)
Statistics	5-fold CV for stability + 1,000-resample bootstrap confidence intervals on test F1
Seed	42 – runs reproduce identical metrics

6. Findings and answers

- RQ1 Classical models work, but spread widely: SVM 0.9459 F1 vs Logistic Regression 0.8585
- RQ2 Ensembles beat individual classifiers, but Voting and Stacking did NOT beat a tuned Random Forest
- RQ3 Yes – all three neural networks are competitive; two sit inside the statistically tied group
- RQ4 Online Boarding and In-flight Wifi Service lead under three independent attribution methods
- RQ5 Performance is tied, so cost and interpretability decide: XGBoost is the practical choice
- DATA Rating 0 means 'not applicable', not 'worst' – 0-raters on wifi are 99.7% satisfied vs ~25% at 1-3

RECOMMENDATION: deploy XGBoost or Random Forest — statistically equal to the neural networks, but faster to train and exactly explainable with SHAP.

SHAP and permutation importance describe model behaviour and association — not causation.